



DUNMAN SECONDARY SCHOOL

CANDIDATE
NAME

CLASS

INDEX
NUMBER

PRELIMINARY EXAMINATION 2024 SECONDARY 4 EXPRESS / 5 NORMAL ACADEMIC

ADDITIONAL MATHEMATICS

Paper 2

4049/02

22 August 2024
2 hours 15 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Calculator Model

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an approved scientific calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 90.

Total Marks

Mathematical Formulae**1. ALGEBRA***Quadratic Equation*

For the equation $ax^2 + bx + c = 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Binomial expansion

$$(a + b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + \binom{n}{r}a^{n-r}b^r + \dots + b^n,$$

where n is a positive integer and $\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)\dots(n-r+1)}{r!}$.

2. TRIGONOMETRY*Identities*

$$\sin^2 A + \cos^2 A = 1$$

$$\sec^2 A = 1 + \tan^2 A$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

Formulae for ΔABC

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Area of } \Delta = \frac{1}{2} ab \sin C$$

Answer **all** the questions.

- 1** Solve the equation $\log_p x + 2 \log_x p = 3$, giving your answer in terms of p , where p is a positive constant. [5]

2 (a) A curve has equation $y = \frac{e^{2x}}{x+2}$, where $x > -2$.

(i) Find $\frac{dy}{dx}$. [2]

(ii) Show that the y -coordinate of the stationary point is $\frac{2}{e^3}$. [2]

(iii) Determine the nature of the stationary point. [2]

- 3 (a) Solve the equation $4^x - 4(2^{x+1}) + 7 = 0$.

[4]

- (b) As people get vaccinated, the number of active cases of people infected with a virus reduces by half each day. Vaccination started at the beginning of 5 February 2021. Given that the number of active cases at the beginning of 5 February 2021 was 59649, find the number of active cases, to the nearest whole number, at the beginning of 12 February 2021.

[2]

- (c) A liquid is allowed to cool after being heated. The temperature, θ °C, of the liquid, t seconds after being removed from the heat, is given by $\theta = 20 + 80e^{-0.01t}$. Collin claimed that the temperature of the liquid after a long time will approach the temperature 20°C. Explain why he is correct. [2]

4 (a) Show that the equation $x^2 + (2 - k)x + k = 3$ has real roots for all real values of k . [3]

(b) Find the smallest value of the integer a for which $ax^2 + 5x + 2$ is positive for all values of x . [3]

(c) Find the values of p and q for which $\{x: x < -3 \text{ or } x > 5\}$ is the solution set of $x^2 - px > q$. [2]

5 The equation of a polynomial is given by $p(x) = x^3 - x^2 + ax + 9$.

(a) Given that $x - 1$ is a factor of $p(x)$, show that $a = -9$. [2]

(b) Solve $x^3 - x^2 - 9x + 9 = 0$. [3]

(c) Hence solve the equation $\cot^3(2y + 30^\circ) - \cot^2(2y + 30^\circ) - 9\cot(2y + 30^\circ) + 9 = 0$
for $0^\circ < y < 180^\circ$. [4]

6 (a) Show that $\frac{d}{dx} \left[x(2x+3)^{\frac{3}{2}} \right] = (5x+3)\sqrt{2x+3}$. [5]

(b) Hence find $\int x\sqrt{2x+3} \, dx$, giving your answer in the form $k(x-1)(2x+3)^{\frac{3}{2}} + c$, where k is a constant to be found and c is a constant of integration which cannot be found. [4]

- 7 A circle, with centre C , passes through the points $P(2, -2)$ and Q .
The point $M(0, 2)$ is the midpoint of the chord PQ and the line CQ is parallel to the x -axis.

(a) Show that C is $(8, 6)$.

[5]

(b) Find the equation of the circle.

[2]

- (c) PR is a diameter of the circle and S is the point $(-1, 13)$.
Determine if the angle PSR is acute or obtuse.

[3]

- 8 A particle travels in a straight line so that, t seconds after passing through O , its velocity, v cm/s, is given by $v = t^2 - 7t + 6$. The particle comes to instantaneous rest, firstly at a point A and then at a point B .
- (a) Find an expression, in terms of t , for the distance, s cm, of the particle, t seconds after passing through O . [1]
- (b) Calculate the distance AB . [4]
- (c) Find the total distance travelled by the particle in the first 6 seconds after passing through O . [2]

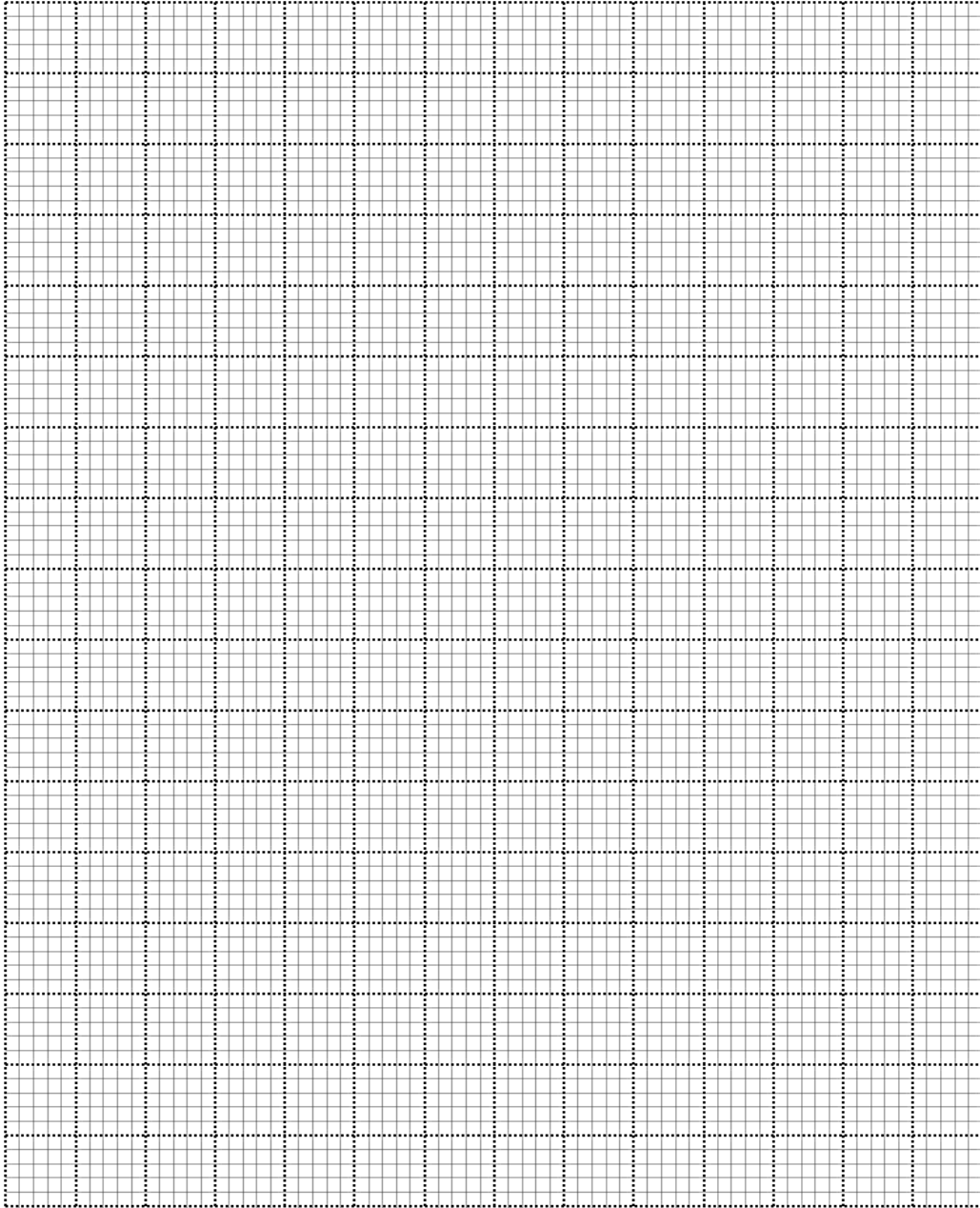
- (d) Given that C is the point at which the particle has zero acceleration, determine, with full working, whether C is nearer to O or to B . [4]

- 9 (a) By considering the general term in the binomial expansion of $\left(x^2 - \frac{2}{x}\right)^7$, explain why every term is dependent on x . [3]

- (b) The first three terms in the expansion of $\left(1 + \frac{x}{6}\right)^n (2 - 3x)^3$ in ascending powers of x , are $8 - 20x + px^2$, where n and p are constants and $n > 0$. Find the values of n and p . [9]

Continuation of working space for Question 9(b).

- 10 (a)** It is known that the variables x and y are related by the equation $y = \frac{p}{x - q}$, where p and q are unknown constants.
- (i)** Express this equation in a form suitable for drawing a straight line graph, and state which variable should be used for each axis. [2]
- (ii)** Explain how the value of p and of q could be determined from this graph. [2]
- (b)** The table shows, to 2 decimal places, the population, P , in millions, of California on January 1st at intervals of 10 years, from 1960 to 2000. The variable x is measured in units of 10 years.
- | | | | | | |
|------|-------|-------|-------|-------|-------|
| Year | 1960 | 1970 | 1980 | 1990 | 2000 |
| x | 0 | 1 | 2 | 3 | 4 |
| P | 15.72 | 19.97 | 23.67 | 29.76 | 33.87 |
- (i)** On the grid on **page 17** plot $\lg P$ against x and draw a straight line graph. [2]



- (iii) Find the gradient of your straight line and hence express P in the form Ab^x , where A and b are constants. [4]

- (iv) The population of California in 2020 was estimated to be 39.37 million. Explain, with mathematical workings, whether the model found in (iii) remains valid. [2]

End of Paper